Comparative Analysis of Premises Valuation Models Using KEEL, RapidMiner, and WEKA

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Abstract. The experiments aimed to compare machine learning algorithms to create models for the valuation of residential premises, implemented in popular data mining systems KEEL, RapidMiner and WEKA, were carried out. Six common methods comprising two neural network algorithms, two decision trees for regression, and linear regression and support vector machine were applied to actual data sets derived from the cadastral system and the registry of real estate transactions. A dozen of commonly used performance measures was applied to evaluate models built by respective algorithms. Some differences between models were observed.

Keywords: machine learning, property valuation, KEEL, RapidMiner, WEKA.

1 Introduction

Sales comparison approach is the most popular way of determining the market value of a property. When applying this method it is necessary to have transaction prices of the properties sold which attributes are similar to the one being appraised. If good comparable sales/purchase transactions are available, then it is possible to obtain reliable estimates. Prior to the evaluation the appraiser must conduct a thorough study of the appraised property using available sources of information such as cadastral systems, transaction registers, performing market analyses, accomplishing on-site inspection. His estimations are usually subjective and are based on his experience and intuition. Automated valuation models (AVMs), devoted to support appraisers' work, are based primarily on multiple regression analysis [8], [11], soft computing and geographic information systems (GIS) [14]. Many intelligent methods have been developed to support appraisers' works: neural networks [13], fuzzy systems [2], case-based reasoning [10], data mining [9] and hybrid approaches [6].

So far the authors have investigated several methods to construct models to assist with real estate appraisal: evolutionary fuzzy systems, neural networks and statistical algorithms using MATLAB and KEEL [4], [5].

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Three non-commercial data mining tools, developed in Java, KEEL [1], RapidMiner [7], and WEKA [12] were chosen to conduct tests. A few common machine learning algorithms including neural networks, decision trees, linear regression methods and support vector machines were included in these tools. Thus we decided to test whether these common algorithms were implemented similarly and allow to generate appraisal models with comparable prediction accuracy. Actual data applied to the experiments with these popular data mining systems came from the cadastral system and the registry of real estate transactions.

2 Cadastral Systems as the Source Base for Model Generation

The concept of data driven models for premises valuation, presented in the paper, was developed on the basis of sales comparison method. It was assumed that whole appraisal area, that means the area of a city or a district, is split into sections (e.g. clusters) of comparable property attributes. The architecture of the proposed system is shown in Fig. 1. The appraiser accesses the system through the internet and chooses an appropriate section and input the values of the attributes of the premises being evaluated into the system, which calculates the output using a given model. The final result as a suggested value of the property is sent back to the appraiser.

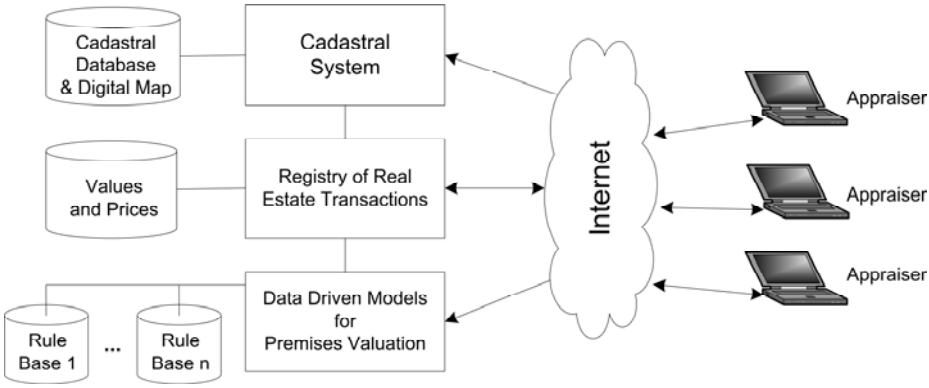

Fig. 1. Information systems to assist with real estate appraisals

Actual data used to generate and learn appraisal models came from the cadastral system and the registry of real estate transactions referring to residential premises sold in one of the big Polish cities at market prices within two years 2001 and 2002. The data set comprised 1098 cases of sales/purchase transactions. Four attributes were pointed out as price drivers: usable area of premises, floor on which premises were located, year of building construction, number of storeys in the building, in turn, price of premises was the output variable.

3 Data Mining Systems Used in Experiments

KEEL (Knowledge Extraction based on Evolutionary Learning) is a software tool to assess evolutionary algorithms for data mining problems including regression, classification, clustering, pattern mining, unsupervised learning, etc. [1]. It comprises evolutionary learning algorithms based on different approaches: Pittsburgh, Michigan, IRL (iterative rule learning), and GCCL (genetic cooperative-competitive learning), as well as the integration of evolutionary learning methods with different pre-processing techniques, allowing it to perform a complete analysis of any learning model.

RapidMiner (RM) is an environment for machine learning and data mining processes [7]. It is open-source, free project implemented in Java. It represents a new approach to design even very complicated problems - a modular operator concept which allows the design of complex nested operator chains for a huge number of learning problems. RM uses XML to describe the operator trees modeling knowledge discovery (KD) processes. RM has flexible operators for data input and output in different file formats. It contains more than 100 learning schemes for regression, classification and clustering tasks.

WEKA (Waikato Environment for Knowledge Analysis) is a non-commercial and open-source project [12]. WEKA contains tools for data pre-processing, classification, regression, clustering, association rules, and visualization. It is also well-suited for developing new machine learning schemes.

4 Regression Algorithms Used in Experiments

In this paper common algorithms for KEEL, RM, and WEKA were chosen. The algorithms represent the same approach to build regression models, but sometimes they have different parameters. Following KEEL, RM, and WEKA algorithms for building, learning and optimizing models were employed to carry out the experiments.

MLP - MultiLayerPerceptron. Algorithm is performed on networks consisting of multiple layers, usually interconnected in a feed-forward way, where each neuron on layer has directed connections to the neurons of the subsequent layer.

RBF - Radial Basis Function Neural Network for Regression Problems. The algorithm is based on feed-forward neural networks with radial activation function on every hidden layer. The output layer represents a weighted sum of hidden neurons signals.

M5P. The algorithm is based on decision trees, however, instead of having values at tree's nodes, it contains a multivariate linear regression model at each node. The input space is divided into cells using training data and their outcomes, then a regression model is built in each cell as a leaf of the tree.

M5R - M5Rules. The algorithm divides the parameter space into areas (subspaces) and builds in each of them a linear regression model. It is based on M5 algorithm. In each iteration a M5 Tree is generated and its best rule is extracted according to a given heuristic. The algorithm terminates when all the examples are covered.

LRM - Linear Regression Model. Algorithm is a standard statistical approach to build a linear model predicting a value of the variable while knowing the values of the other variables. It uses the least mean square method in order to adjust the parameters of the linear model/function.

SVM - NU-Support Vector Machine. Algorithm constructs support vectors in high-dimensional feature space. Then, hyperplane with the maximal margin is constructed. Kernel function is used to transform the data, which augments the dimensionality of the data. This augmentation provokes that the data can be separated with an hyperplane with much higher probability, and establish a minimal prediction probability error measure.

5 Plan of Experiments

The main goal of our study was to compare six algorithms for regression, which are common for KEEL, RM and WEKA. There were: multilayer perceptron, radial-basis-function networks, two types of model trees, linear regression, and support vector machine, and they are listed in Table 1. They were arranged in 4 groups: neural networks for regression (NNR), decision tree for regression (DTR), statistical regression model (SRM), and support vector machine (SVM).

Table 1. Machine learning algorithms used in study

Type	Code	KEEL name	RapidMiner name	WEKA name
NNR	MLP	Regr-MLPerceptronConj-Grad	W-MultilayerPerceptron	MultilayerPerceptron
	RBF	Regr-RBFN	W-RBFNetwork	RBFNetwork
DTR	M5P	Regr-M5	W-M5P	M5P
	M5R	Regr-M5Rules	W-M5Rules	M5Rules
SRM	LRM	Regr-LinearLMS	LinearRegression	LinearRegression
SVM	SVM	Regr-NU_SVR	LibSVM	LibSVM

Optimal parameters were selected for every algorithm to get the best result for the dataset by means of trial and error method. Having determined the best parameters of respective algorithms, final experiments were carried out in order to compare predictive accuracy of models created using all six selected algorithms in KEEL, RM, and WEKA. Schema of the experiments is depicted in Figure 2. All the experiments were run for our set of data using 10-fold cross validation. In order to obtain comparable results, the normalization of data was performed using the min-max approach. As fitness function the mean square error (MSE) programmed in KEEL, and root mean square error (RMSE) implemented in WEKA and RM were used (MSE = $RMSE^2$). Nonparametric Wilcoxon signed-rank test was employed to evaluate the outcome. A dozen of commonly used performance measures [3], [12] were applied to evaluate models built by respective algorithms used in our study. These measures are listed in Table 2 and expressed in the form of following formulas below, where y_i denotes actual price and $\hat{y}_i$ – predicted price of i-th case, *avg(v), var(v), std(v)* – average, variance, and standard deviation of variables $v_1, v_2, \ldots, v_N$, respectively and N – number of cases in the testing set.

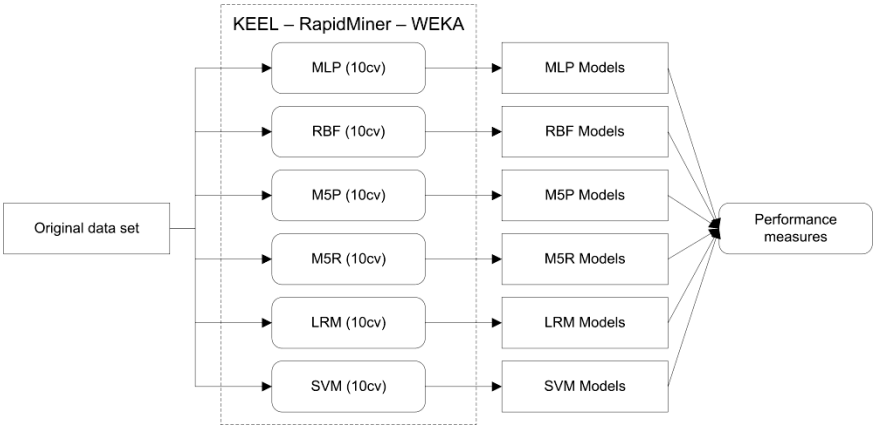

Fig. 2. Schema of the experiments with KEEL, RapidMiner, and WEKA

Table 2. Performance measures used in study

Denot.	Description	Dimen-sion	Min value	Max value	Desirable outcome	No. of form.
MSE	Mean squared error	d^2	0	∞	min	1
RMSE	Root mean squared error	d	0	∞	min	2
RSE	Relative squared error	no	0	∞	min	3
RRSE	Root relative squared error	no	0	∞	min	4
MAE	Mean absolute error	d	0	∞	min	5
RAE	Relative absolute error	no	0	∞	min	6
MAPE	Mean absolute percent. error	%	0	∞	min	7
NDEI	Non-dimensional error index	no	0	∞	min	8
r	Linear correlation coefficient	no	-1	1	close to 1	9
R^2	Coefficient of determination	%	0	∞	close to 100%	10
var(AE)	Variance of absolute errors	d^2	0	∞	min	11
var(APE)	Variance of absolute percentage errors	no	0	∞	min	12

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{1}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{2}$$

$$RSE = \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - avg(y))^2} \tag{3}$$

$$RRSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - avg(y))^2}} \tag{4}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{5}$$

$$RAE = \frac{\sum_{i=1}^{N}|y_i - \hat{y}_i|}{\sum_{i=1}^{N}|y_i - avg(y)|} \tag{6}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{y_i} * 100\% \tag{7}$$

$$NDEI = \frac{RMSE}{std(y)} \tag{8}$$

$$r = \frac{\sum_{i=1}^{N}(y_i - avg(y))\,(\hat{y}_i - avg(\hat{y}))}{\sqrt{\sum_{i=1}^{N}(y_i - avg(y))^2}\sqrt{\sum_{i=1}^{N}(\hat{y}_i - avg(\hat{y}))^2}} \tag{9}$$

$$R^2 = \frac{\sum_{i=1}^{N}(\hat{y}_i - avg(y))^2}{\sum_{i=1}^{N}(y_i - avg(y))^2} * 100\% \tag{10}$$

$$var(AE) = var(|y - \hat{y}|) \tag{11}$$

$$var(APE) = var(\frac{|y - \hat{y}|}{y}) \tag{12}$$

6 Results of Experiments

The performance of the models built by all six algorithms for respective measures and for KEEL, RM, and WEKA systems was presented in Fig. 3-14. For clarity, all measures were calculated for normalized values of output variables except for MAPE, where in order to avoid the division by zero, actual and predicted prices had to be denormalized. It can be observed that, some measures, especially those based on square errors reveal similar relationships between model performance. Most of the models provided similar values of error measures, besides the one created by MLP algorithm implemented in RapidMiner. Its worst performance can be seen particularly in Figures 8, 9, 11, and 13.

Fig. 9 depicts that the values of MAPE range from 16.2% to 19.3%, except for MLP in RapidMiner with 25.3%, what is a fairly good result, especially when you take into account, that no all price drivers were available in our sources of experimental data.

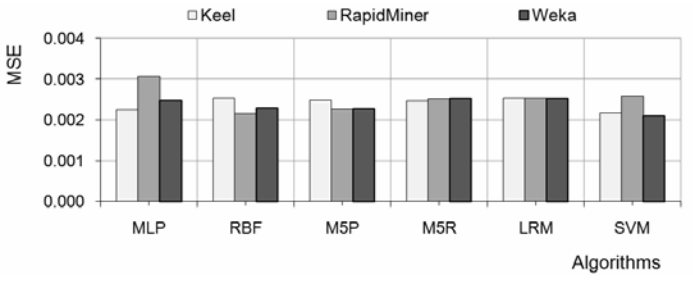

Fig. 3. Comparison of MSE values

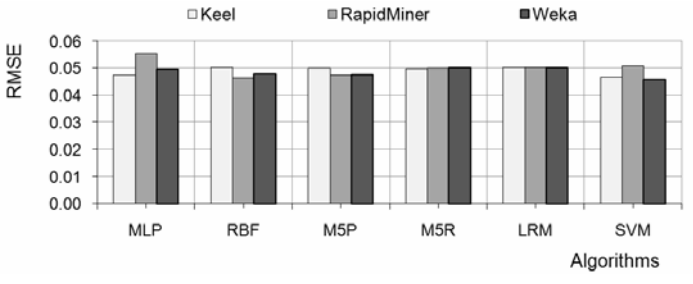

Fig. 4. Comparison of RMSE values

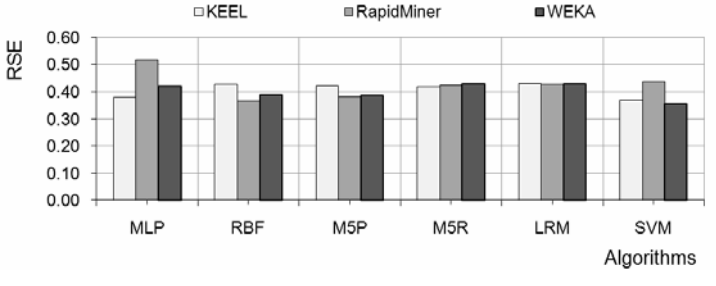

Fig. 5. Comparison of RSE values

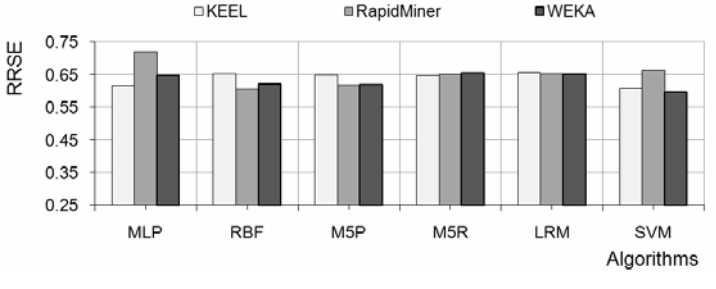

Fig. 6. Comparison of RRSE values

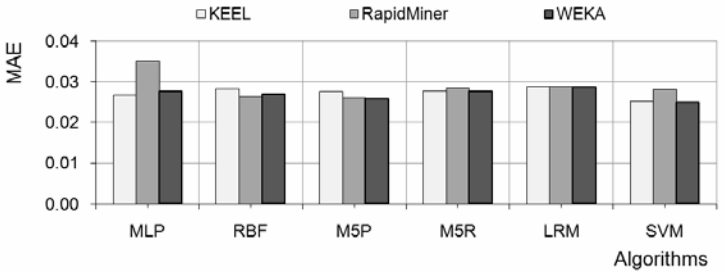

Fig. 7. Comparison of MAE values

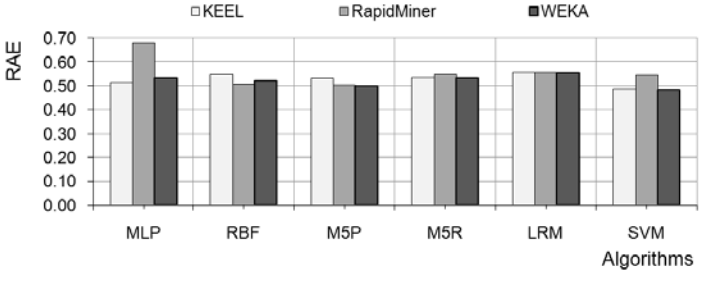

Fig. 8. Comparison of RAE values

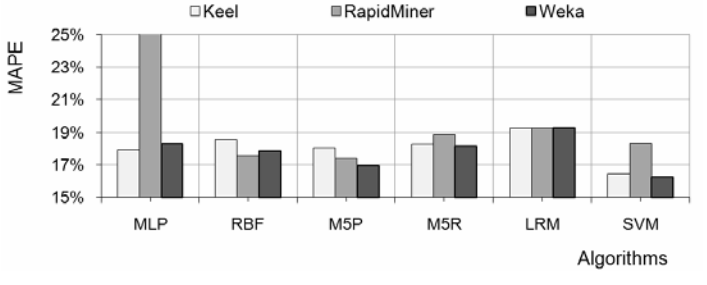

Fig. 9. Comparison of MAPE values

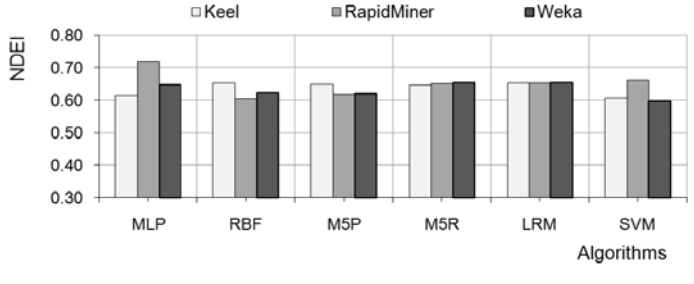

Fig. 10. Comparison of NDEI values

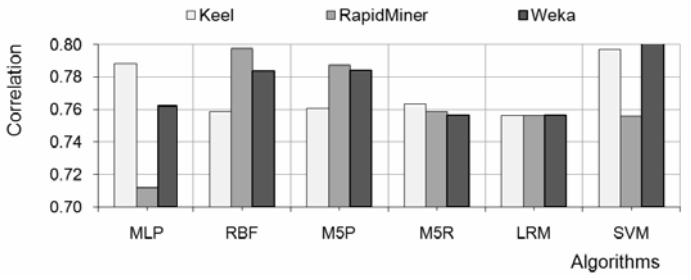

Fig. 11. Correlation between predicted and actual prices

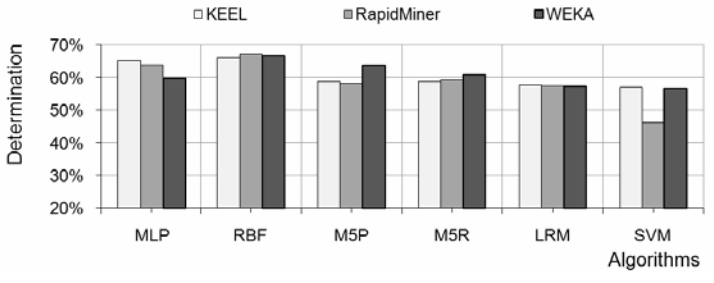

Fig. 12. Comparison of R^2 - determination coefficient values

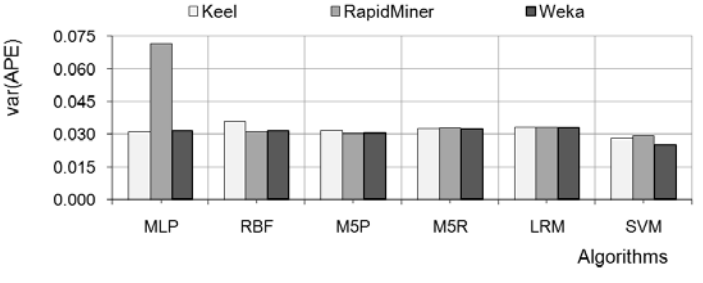

Fig. 13. Variance of absolute percentage errors - Var(APE)

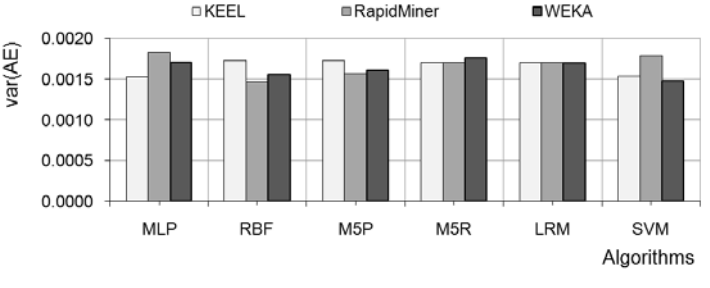

Fig. 14. Variance of absolute errors - Var(AE)

High correlation between actual and predicted prices for each model, ranging from 71.2% to 80.4%, was shown in Fig. 11. In turn, the coefficients of determination, presented in Fig. 12, indicate that from 46.2% to 67.2% of total variation in the dependent variable (prices) is accounted for by the models. This can be explained that data derived from the cadastral system and the register of property values and prices cover only some part of potential price drivers.

The nonparametric Wilcoxon signed-rank tests were carried out for two commonly used measures: MSE and MAPE. The results are shown in Tables 3 and 4, where a triple in each cell, eg <NNN>, reflects the outcome for a given pair of models and for KEEL, RM, and WEKA respectively. N - denotes that there are no differences in mean values of respective errors, and Y - indicates that there are statistically significant differences between particular performance measures. For clarity Tables 3 and 4 were presented in form of symmetric matrices.

Almost in all cases but one SVM revealed significantly better performance than other algorithms, whereas LRM turned out to be worse.

Table 3. Results of Wilcoxon signed-rank test for squared errors comprised by MSE

Model	MLP	RBF	M5P	M5R	LRM	SVM
MLP	-	YYY	YYY	YYN	YYY	YYY
RBF	YYY	-	NNY	NYN	YYY	YYY
M5P	YYY	NNY	-	NYY	YYY	YYN
M5R	YYN	NYN	NYY	-	YNY	YYY
LRM	YYY	YYY	YYY	YNY	-	YYY
SVM	YYY	YYY	YYN	YYY	YYY	-

Table 4. Results of Wilcoxon test for absolute percentage errors comprised by MAPE

Model	MLP	RBF	M5P	M5R	LRM	SVM
MLP	-	NYY	NYY	NYN	YYY	YYY
RBF	NYY	-	NNY	NYN	YYY	YYY
M5P	NYY	NNY	-	NYY	YYY	YYN
M5R	NYN	NYN	NYY	-	YNY	YYY
LRM	YYY	YYY	YYY	YNY	-	YYY
SVM	YYY	YYY	YYN	YYY	YYY	-

Due to the non-decisive results of statistical tests for other algorithms, rank positions of individual algorithms were determined for each measure (see Table 5). It can be noticed that highest rank positions gained SVM, RBF, and M5P algorithms and the lowest LRM, M5R, and MLP. There are differences in rankings made on the basis of the performance of algorithms within respective data mining systems.

In order to find out to what extent the three data mining systems produce uniform models for the same algorithms the nonparametric Wilcoxon signed-rank tests were carried out for three measures: MSE, MAE and MAPE. The results are shown in Table 6, where a triple in each cell, eg <NNN>, reflects outcome for each pair of models created by KEEL-RM, KEEL-WEKA, and RM-WEKA respectively. N - denotes that there are no differences in mean values of respective errors, and Y - indicates that there are statistically significant differences between particular performance measures. Some general conclusions can be drawn when analysing Table 6. For LRM models there is

Table 5. Rank positions of algorithms with respect to performance measures (1 means the best)

Measure	Tool	MLP	RBF	M5P	M5R	LRM	SVM
MSE	KEEL	2	5	4	3	6	1
MSE	RM	6	1	2	3	4	5
MSE	WEKA	4	3	2	6	5	1
RMSE	KEEL	2	5	4	3	6	1
RMSE	RM	6	1	2	3	4	5
RMSE	WEKA	4	3	2	6	5	1
RSE	KEEL	3	1	5	4	6	2
RSE	RM	6	4	1	2	3	5
RSE	WEKA	4	3	2	6	5	1
RRSE	KEEL	3	1	5	4	6	2
RRSE	RM	6	4	1	2	3	5
RRSE	WEKA	4	3	2	6	5	1
MAE	KEEL	3	2	4	5	6	1
MAE	RM	6	3	1	4	5	2
MAE	WEKA	5	3	2	4	6	1
RAE	KEEL	3	2	4	5	6	1
RAE	RM	6	3	1	4	5	2
RAE	WEKA	5	3	2	4	6	1
r	KEEL	3	1	5	4	6	2
r	RM	6	2	1	3	4	5
r	WEKA	4	3	2	5	6	1
MAPE	KEEL	3	2	4	5	6	1
MAPE	RM	6	3	1	4	5	2
MAPE	WEKA	5	3	2	4	6	1
NDEI	KEEL	2	5	4	3	6	1
NDEI	RM	6	1	2	3	4	5
NDEI	WEKA	4	3	2	6	5	1
r	KEEL	3	1	5	4	6	2
r	RM	6	2	1	3	4	5
r	WEKA	4	3	2	5	6	1
R^2	KEEL	5	6	3	4	2	1
R^2	RM	6	5	3	4	2	1
R^2	WEKA	3	6	5	4	2	1
Var(AE)	KEEL	2	1	6	4	5	3
Var(AE)	RM	6	4	1	3	2	5
Var(AE)	WEKA	5	2	3	6	4	1
Var(APE)	KEEL	3	2	4	5	6	1
Var(APE)	RM	6	5	2	3	4	1
Var(APE)	WEKA	4	3	2	5	6	1
median		4.00	3.00	2.00	4.00	6.00	1.00
average		4.32	2.58	2.63	4.16	5.42	1.89
min		2	1	1	3	4	1
max		6	5	5	6	6	5

Table 6. Results of Wilcoxon test for common algorithms in KEEL, RM, and WEKA

Measure	MLP	RBF	M5P	M5R	LRM	SVM
MSE	YNY	YNN	YNN	YNN	NNN	YNY
MAE	YNY	YNN	YNN	YNN	NNN	YNY
MAPE	YNY	YNN	YNN	YNN	NNN	YNY

no significant difference in performance, the same applies to all models built by KEEL and WEKA. For each model created by means of KEEL and RapidMiner there are significant differences in prediction accuracy.

7 Conclusions and Future Work

The experiments aimed to compare machine learning algorithms to create models for the valuation of residential premises, implemented in popular data mining systems KEEL, RapidMiner and WEKA, were carried out. Six common methods were applied to actual data sets derived from the cadastral systems, therefore it was naturally to expect that the same algorithms implemented in respective systems will produce similar results. However, this was true only for KEEL and WEKA systems, and for linear regression method. For each algorithm there were significant differences between KEEL and RapidMiner.

Some performance measures provide the same distinction abilities of respective models, thus it can be concluded that in order to compare a number of models it is not necessary to employ all measures, but the representatives of different groups.

MAPE obtained in all tests ranged from 16% do 25%. This can be explained that data derived from the cadastral system and the register of property values and prices can cover only some part of potential price drivers. Physical condition of the premises and their building, their equipment and facilities, the neighbourhood of the building, the location in a given part of a city should also be taken into account, moreover overall subjective assessment after inspection in site should be done. Therefore we intend to test data obtained from public registers and then supplemented by experts conducting on-site inspections and evaluating more aspects of properties being appraised. Moreover further investigations of multiple models comprising ensembles using bagging and boosting techniques is planned.

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